

## An Analysis of Mean squared error

Mean squared error

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

## Mean squared error -- Part 1

In other words, the MSE is the mean  $\left(\frac{1}{n} \sum_{i=1}^n \left(Y_i - \hat{Y}_i\right)^2\right)$  of the squares of the errors  $\left(Y_i - \hat{Y}_i\right)^2$ . This is an easily computable quantity for a particular sample (and hence is sample-dependent). In matrix notation,

## Mean squared error -- Part 2

where  $\sigma^2$  is the population variance. For a Gaussian distribution this is the best unbiased estimator of the population mean, that is the one with the lowest MSE (and hence variance) among all unbiased estimators. One can check that the MSE above equals the inverse of the Fisher information (see Cramér–Rao bound). But the same sample mean is not the best estimator of the population mean, say, for a uniform distribution.

## Mean squared error -- Part 3

where  $e_i = Y_i - \hat{Y}_i$  and  $\mathbf{e}$  is a  $n \times 1$  column vector. The MSE can also be computed on  $q$  data points that were not used in estimating the model, either because they were held back for this purpose, or because these data have been newly obtained. Within this process, known as cross-validation, the MSE is often called the test MSE, and is computed as

## Mean squared error -- Part 4

$$\begin{aligned} \text{MSE} \blacksquare (S_{a2}) &= E \blacksquare [(n-1) a S_{n-1-2-\sigma 2})^2] = E \blacksquare [(n-1)^2 a^2 S_{n-1-4-2(n-1) a S_{n-1-2}} \sigma^2 + \sigma^4] = (n-1)^2 a^2 E \blacksquare [S_{n-1-4}] - 2(n-1) a E \blacksquare [S_{n-1-2}] \sigma^2 + \sigma^4 = \\ &= (n-1)^2 a^2 E \blacksquare [S_{n-1-4}] - 2(n-1) a \sigma^4 + \sigma^4 E \blacksquare [S_{n-1-2}] = \sigma^2 = (n-1)^2 a^2 (\gamma^2 n + n + 1) \sigma^4 - 2(n-1) a \sigma^4 + \sigma^4 E \blacksquare [S_{n-1-4}] = \text{MSE} \blacksquare (S_{n-1-2}) + \sigma^4 = n-1 \\ &= n a^2 ((n-1) \gamma^2 + n^2 + n) \sigma^4 - 2(n-1) a \sigma^4 + \sigma^4 \end{aligned}$$

$$\begin{aligned} & \left\{ \frac{(n-1)^2}{a^2} \right\} \operatorname{operatorname {E}} \\ & \left[ S_{n-1}^4 \right] - 2 \left( \frac{ \\ \left\{ \frac{(n-1)^2}{a^2} \right\} \operatorname{operatorname {E}} \\ & \left[ S_{n-1}^2 \right] \sigma^2 + \sigma^4 \right) \frac{ \\ \left\{ \frac{(n-1)^2}{a^2} \right\} \operatorname{operatorname {E}} \\ & \left[ S_{n-1}^4 \right] - 2 \left( \frac{ \\ \left\{ \frac{(n-1)^2}{a^2} \right\} \operatorname{operatorname {E}} \\ & \left[ S_{n-1}^2 \right] = \sigma^2 \\ & \left\{ \frac{(n-1)^2}{a^2} \right\} \left( \frac{ \\ \left\{ \frac{(n-1)^2}{a^2} \right\} \operatorname{operatorname {E}} \\ & \left[ S_{n-1}^4 \right] = \operatorname{operatorname {MSE}} \\ & (S_{n-1}^2) + \sigma^4 \\ & \left\{ \frac{(n-1)^2}{a^2} \right\} \left( \frac{ \\ \left\{ \frac{(n-1)^2}{a^2} \right\} \operatorname{operatorname {E}} \\ & \left[ S_{n-1}^4 \right] - 2 \left( \frac{ \\ \left\{ \frac{(n-1)^2}{a^2} \right\} \operatorname{operatorname {E}} \\ & \left[ S_{n-1}^2 \right] + \sigma^4 \right) \end{aligned}$$

## Mean squared error -- Part 5

Criticism The use of mean squared error without question has been criticized by the decision theorist James Berger. Mean squared error is the negative of the expected value of one specific utility function, the quadratic utility function, which may not be the appropriate utility function to use under a given set of circumstances. There are, however, some scenarios where mean squared error can serve as a good approximation to a loss function occurring naturally in an application. Like variance, mean squared error has the disadvantage of heavily weighting outliers. This is a result of the squaring of each term, which effectively weights large errors more heavily than small ones. This property, undesirable in many applications, has led researchers to use alternatives such as the mean absolute error, or those based on the median.

## Mean squared error -- Part 6

$$S_{a-1}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

## Mean squared error -- Part 7

This definition depends on the unknown parameter, therefore the MSE is a priori property of an estimator. The MSE could be a function of unknown parameters, in which case any estimator of the MSE based on estimates of these parameters would be a function of the data (and thus a random variable). If the estimator  $\hat{\theta}$  is derived as a sample statistic and is used to estimate some population parameter, then the expectation is with respect to the sampling distribution of the

# An Analysis of Mean squared error

*Mean squared error*

sample statistic. The MSE can be written as the sum of the variance of the estimator and the squared bias of the estimator, providing a useful way to calculate the MSE and implying that in the case of unbiased estimators, the MSE and variance are equivalent.

## Mean squared error -- Part 8

---

Applications Minimizing MSE is a key criterion in selecting estimators; see minimum mean-square error. Among unbiased estimators, minimizing the MSE is equivalent to minimizing the variance, and the estimator that does this is the minimum variance unbiased estimator. However, a biased estimator may have lower MSE; see estimator bias. In statistical modelling the MSE can represent the difference between the actual observations and the observation values predicted by the model. In this context, it is used to determine the extent to which the model fits the data as well as whether removing some explanatory variables is possible without significantly harming the model's predictive ability. In forecasting and prediction, the Brier score is a measure of forecast skill based on MSE.

## Mean squared error -- Part 9

---

$$\begin{aligned}\text{MSE}(\hat{\theta}) &= E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta} - \theta) + (E[\hat{\theta} - \theta])^2 \\ &= \text{Var}(\hat{\theta}) + \text{Bias}^2(\hat{\theta}, \theta) \\\end{aligned}$$